

Desc Forged Equal Tee ASTM A/SA105 NPT Threaded,2" 3000#

CERTIFIED MILL TEST REPORT



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BONNEY FORGE CORPORATION

P.O. BOX 330 • 14496 CROGHAN PIKE • MOUNT UNION, PA 17066-0330

(814) 542-2545 • (800) 345-7546 • FAX (814) 542-4906

www.bonneyforge.com

CUSTOMER: EDGEN MURRAY CORPORATION

DATE 02/13/2025

CUSTOMER

BONNEY ORDER NO. B000358755

ORDER NO.: DIS-313586

SHIP TO: EDGEN MURRAY CORPORATION

8445 MILLER RD 2

HOUSTON TX 77049

United States

ITEM	QUANTITY	LOT NO.	GRADE OR SPECIFICATION NO. CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:
1	25	51763	1 3000 A105 45EL T SA/A105 Al 0.028 C 0.180 Cb 0.003 Co 0.006 Cr 0.120 Cu 0.120 Mn 1.050 Mo 0.030 Ni 0.060 P 0.006 S 0.018 Si 0.230 Sn 0.004 V 0.035 CE(Long Formula) = 0.40 T/S(PSI) 81,773 Y/S(PSI) 55,902 EL(%) 32.85 RA(%) 62.52 Brinell Hardness 159 BHN 153 BHN
2	25	51826	1 3000 A105 45EL S SA/A105 Al 0.025 C 0.200 Co 0.002 Cr 0.120 Cu 0.030 Mn 1.080 Mo 0.004 Nb 0.011 Ni 0.020 P 0.006 S 0.023 Si 0.220 V 0.004 CE(Long Formula) = 0.41 T/S(PSI) 82,518 Y/S(PSI) 51,753 EL(%) 30.65 RA(%) 59.32 Brinell Hardness 165 BHN 169 BHN
4	20	51913	11/4 3000 A105 90EL T SA/A105 Al 0.022 C 0.200 Co 0.003 Cr 0.110 Cu 0.040 Mn 1.100 Mo 0.006 Nb 0.010 Ni 0.030 P 0.006 S 0.023 Si 0.200 V 0.004 CE(Long Formula) = 0.41 T/S(PSI) 80,484 Y/S(PSI) 52,124 EL(%) 33.90 RA(%) 63.62 Brinell Hardness 147 BHN 144 BHN
5	25	51948	11/4 3000 A105 90EL S SA/A105 Al 0.021 C 0.200 Co 0.003 Cr 0.130 Cu 0.040 Mn 1.070 Mo 0.003 Nb 0.009 Ni 0.010 P 0.006 S 0.020 Si 0.220 V 0.003 CE(Long Formula) = 0.41 T/S(PSI) 81,675 Y/S(PSI) 50,147 EL(%) 32.10 RA(%) 63.47 Brinell Hardness 144 BHN 144 BHN

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by

Paige Smith

Paige Smith
Quality Assurance Representative

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
6	60	51917	11/2 3000 A105 90EL T	SA/A105
Al 0.028 C 0.200 Co 0.003 Cr 0.090 Cu 0.050 Mn 1.040 Mo 0.024 Nb 0.010				
Ni 0.030 P 0.005 S 0.021 Si 0.230 V 0.004				
CE(Long Formula) = 0.40				
T/S(PSI) 80,160 Y/S(PSI) 51,354 EL(%) 34.10 RA(%) 60.78				
Brinell Hardness 150 BHN 147 BHN				
7	175	51913	11/2 3000 A105 90EL S	SA/A105
Al 0.022 C 0.200 Co 0.003 Cr 0.110 Cu 0.040 Mn 1.100 Mo 0.006 Nb 0.010				
Ni 0.030 P 0.006 S 0.023 Si 0.200 V 0.004				
CE(Long Formula) = 0.41				
T/S(PSI) 80,484 Y/S(PSI) 52,124 EL(%) 33.90 RA(%) 63.62				
Brinell Hardness 147 BHN 144 BHN				
8	10	51889	11/2 6000 A105 90EL S	SA/A105
Al 0.023 C 0.190 Co 0.002 Cr 0.120 Cu 0.020 Mn 1.060 Mo 0.001 Nb 0.012				
Ni 0.010 P 0.004 S 0.021 Si 0.200 V 0.004				
CE(Long Formula) = 0.39				
T/S(PSI) 76,386 Y/S(PSI) 49,280 EL(%) 32.40 RA(%) 61.83				
Brinell Hardness 144 BHN 144 BHN				
10	24	51760	1 6000 A105 90EL T	SA/A105
Al 0.028 C 0.180 Co 0.006 Cr 0.140 Cu 0.160 Mn 1.090 Mo 0.040 Nb 0.003				
Ni 0.080 P 0.007 S 0.024 Si 0.270 V 0.036				
CE(Long Formula) = 0.42				
T/S(PSI) 80,292 Y/S(PSI) 55,211 EL(%) 32.00 RA(%) 61.40				
Brinell Hardness 153 BHN 153 BHN				

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
11	63	51897	2 3000 A105 90EL T	SA/A105
Al 0.025 C 0.170 Cb 0.000 Co 0.006 Cr 0.110 Cu 0.160 Mn 1.160 Mo 0.040				
Ni 0.060 P 0.008 S 0.022 Si 0.250 V 0.034				
CE(Long Formula) = 0.41				
T/S(PSI) 81,887 Y/S(PSI) 51,134 EL(%) 32.55 RA(%) 65.51				
Brinell Hardness 156 BHN 153 BHN				
12	18	51811	2 6000 A105 90EL S	SA/A105
Al 0.027 C 0.180 Co 0.007 Cr 0.170 Cu 0.330 Mn 1.030 Mo 0.010 Nb 0.003				
Ni 0.070 P 0.013 S 0.022 Si 0.211 V 0.021				
CE(Long Formula) = 0.42				
T/S(PSI) 80,343 Y/S(PSI) 52,262 EL(%) 33.50 RA(%) 70.71				
Brinell Hardness 141 BHN 144 BHN				
16	25	D993	1 3000 A105 CAP S	SA/A105
Al 0.029 C 0.180 Co 0.006 Cr 0.070 Cu 0.260 Mn 1.040 Mo 0.020 Nb 0.001				
Ni 0.080 P 0.004 S 0.025 Si 0.215 V 0.025				
CE(Long Formula) = 0.40				
T/S(PSI) 73,500 Y/S(PSI) 49,600 EL(%) 32.00 RA(%) 38.00				
Brinell Hardness 163 BHN 163 BHN				
17	25	J101	1 6000 A105 CAP T	SA/A105
Al 0.034 C 0.190 Co 0.006 Cr 0.160 Cu 0.250 Mn 0.980 Mo 0.020 Nb 0.002				
Ni 0.090 P 0.013 S 0.016 Si 0.200 V 0.019				
CE(Long Formula) = 0.42				
T/S(PSI) 79,300 Y/S(PSI) 48,300 EL(%) 27.00 RA(%) 48.00				
Brinell Hardness 161 BHN 159 BHN				

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
19	25	D750	3/8 3000 A105 COUP T	SA/A105
Al 0.020 C 0.200 Co 0.010 Cr 0.130 Cu 0.100 Mn 1.000 Mo 0.010 Nb 0.002				
Ni 0.040 P 0.007 S 0.020 Si 0.240 V 0.003				
CE(Long Formula) = 0.40				
T/S(PSI) 76,100 Y/S(PSI) 51,200 EL(%) 32.00 RA(%) 65.00				
Brinell Hardness 149 BHN 148 BHN				
20	25	J131	11/4 3000 A105 COUP T	SA/A105
Al 0.026 C 0.200 Co 0.002 Cr 0.120 Cu 0.030 Mn 1.030 Mo 0.001 Nb 0.018				
Ni 0.010 P 0.007 S 0.021 Si 0.200 V 0.004				
CE(Long Formula) = 0.40				
T/S(PSI) 78,800 Y/S(PSI) 46,800 EL(%) 32.50 RA(%) 57.20				
Brinell Hardness 156 BHN 160 BHN				
22	25	J103	1 3000 A105 COUP T	SA/A105
Al 0.029 C 0.190 Co 0.006 Cr 0.080 Cu 0.260 Mn 0.980 Mo 0.020 Nb 0.002				
Ni 0.070 P 0.006 S 0.018 Si 0.211 V 0.024				
CE(Long Formula) = 0.40				
T/S(PSI) 76,800 Y/S(PSI) 54,700 EL(%) 35.00 RA(%) 38.00				
Brinell Hardness 167 BHN 164 BHN				
23	48	J107	2 3000 A105 COUP T	SA/A105
Al 0.027 C 0.170 Co 0.006 Cr 0.150 Cu 0.250 Mn 1.000 Mo 0.020 Nb 0.002				
Ni 0.080 P 0.009 S 0.019 Si 0.221 V 0.028				
CE(Long Formula) = 0.40				
T/S(PSI) 79,200 Y/S(PSI) 51,500 EL(%) 24.00 RA(%) 36.00				
Brinell Hardness 166 BHN 164 BHN				

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
		J147	2 3000 A105 COUP T	SA/A105
			Al 0.031 C 0.200 Co 0.005 Cr 0.130 Cu 0.270 Mn 1.000 Mo 0.010 Nb 0.003	
			Ni 0.060 P 0.008 S 0.022 Si 0.220 V 0.020	
			CE(Long Formula) = 0.42	
			T/S(PSI) 76,500 Y/S(PSI) 52,200 EL(%) 26.00 RA(%) 48.00	
			Brinell Hardness 156 BHN 156 BHN	
24	25	D452	2 3000 A105 COUP S	SA/A105
			C 0.220 Cb 0.009 Co 0.004 Cr 0.100 Cu 0.070 Mn 1.090 Mo 0.005 N 0.004	
			Ni 0.040 P 0.006 S 0.024 Si 0.200 Ti 0.001 V 0.004	
			CE(Long Formula) = 0.43	
			T/S(PSI) 79,000 Y/S(PSI) 54,600 EL(%) 28.50 RA(%) 55.50	
			Brinell Hardness 158 BHN 160 BHN	
25	12	D912	2 6000 A105 COUP S	SA/A105
			Al 0.024 C 0.180 Cb 0.000 Co 0.004 Cr 0.110 Cu 0.130 Mn 1.120 Mo 0.030	
			Ni 0.070 P 0.008 S 0.022 Si 0.240 V 0.032	
			CE(Long Formula) = 0.41	
			T/S(PSI) 79,100 Y/S(PSI) 54,200 EL(%) 29.80 RA(%) 63.40	
			Brinell Hardness 160 BHN 157 BHN	
28	50	D741	1/4X 1/8 3M/6M A105 HXBU T	SA/A105
			Al 0.020 C 0.200 Co 0.010 Cr 0.130 Cu 0.100 Mn 1.000 Mo 0.010 Nb 0.002	
			Ni 0.040 P 0.007 S 0.020 Si 0.240 V 0.003	
			CE(Long Formula) = 0.40	
			T/S(PSI) 75,900 Y/S(PSI) 52,600 EL(%) 33.00 RA(%) 71.00	
			Brinell Hardness 146 BHN 137 BHN	

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
30	50	D914	3/4X 1/4 3M/6M A105 HXBU T	SA/A105
Al 0.026 C 0.200 Co 0.012 Cr 0.070 Cu 0.300 Mn 1.050 Mo 0.020 Nb 0.005				
Ni 0.090 P 0.009 S 0.023 Si 0.233 V 0.022				
CE(Long Formula) = 0.42				
T/S(PSI) 77,700 Y/S(PSI) 54,500 EL(%) 35.00 RA(%) 35.00				
Brinell Hardness 167 BHN 169 BHN				
31	25	D902	1 1/2X 1/4 3M/6M A105 HXBU T	SA/A105
Al 0.024 C 0.210 Co 0.002 Cr 0.120 Cu 0.030 Mn 1.050 Mo 0.004 Nb 0.011				
Ni 0.020 P 0.007 S 0.022 Si 0.220 V 0.003				
CE(Long Formula) = 0.41				
T/S(PSI) 78,500 Y/S(PSI) 46,700 EL(%) 29.00 RA(%) 51.60				
Brinell Hardness 164 BHN 164 BHN				
32	25	D995	1 X 1/4 3M/6M A105 HXBU T	SA/A105
Al 0.028 C 0.200 Co 0.006 Cr 0.080 Cu 0.260 Mn 0.980 Mo 0.020 Nb 0.002				
Ni 0.070 P 0.006 S 0.019 Si 0.210 V 0.023				
CE(Long Formula) = 0.41				
T/S(PSI) 78,400 Y/S(PSI) 54,800 EL(%) 32.00 RA(%) 39.00				
Brinell Hardness 160 BHN 154 BHN				
33	50	D994	1 X 1/2 3M/6M A105 HXBU T	SA/A105
Al 0.029 C 0.190 Co 0.006 Cr 0.080 Cu 0.260 Mn 0.980 Mo 0.020 Nb 0.002				
Ni 0.070 P 0.006 S 0.018 Si 0.211 V 0.024				
CE(Long Formula) = 0.40				
T/S(PSI) 77,200 Y/S(PSI) 54,300 EL(%) 35.00 RA(%) 44.00				
Brinell Hardness 163 BHN 164 BHN				

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
34	75	D994	1 X 3/4 3M/6M A105 HXBU T	SA/A105
			Al 0.029 C 0.190 Co 0.006 Cr 0.080 Cu 0.260 Mn 0.980 Mo 0.020 Nb 0.002	
			Ni 0.070 P 0.006 S 0.018 Si 0.211 V 0.024	
			CE(Long Formula) = 0.40	
			T/S(PSI) 77,200 Y/S(PSI) 54,300 EL(%) 35.00 RA(%) 44.00	
		J124	1 X 3/4 3M/6M A105 HXBU T	SA/A105
			Al 0.026 C 0.200 Co 0.003 Cr 0.120 Cu 0.030 Mn 1.070 Mo 0.010 Nb 0.012	
			Ni 0.030 P 0.005 S 0.022 Si 0.210 V 0.004	
			CE(Long Formula) = 0.41	
			T/S(PSI) 80,000 Y/S(PSI) 50,400 EL(%) 35.00 RA(%) 61.60	
35	25	D959	2 X 1/2 3M/6M A105 HXBU T	SA/A105
			Al 0.040 C 0.190 Co 0.006 Cr 0.210 Cu 0.260 Mn 1.020 Mo 0.020 Nb 0.003	
			Ni 0.060 P 0.013 S 0.020 Si 0.224 V 0.019	
			CE(Long Formula) = 0.43	
			T/S(PSI) 81,500 Y/S(PSI) 50,800 EL(%) 30.00 RA(%) 64.00	
37	50	D940	1/4 3M/6M A105 HXPL T	SA/A105
			C 0.190 Cb 0.000 Co 0.000 Cr 0.060 Cu 0.170 Mn 0.740 Mo 0.020 Ni 0.080	
			P 0.006 S 0.013 Si 0.250 V 0.003	
			CE(Long Formula) = 0.35	
			T/S(PSI) 77,700 Y/S(PSI) 74,900 EL(%) 23.00 RA(%) 63.00	
			Brinell Hardness 156 BHN 165 BHN	

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38	25	D858	3/8 3M/6M A105 HXPL T	SA/A105
C 0.160 Cb 0.000 Co 0.000 Cr 0.070 Cu 0.090 Mn 0.740 Mo 0.020 Ni 0.050				
P 0.006 S 0.008 Si 0.170 V 0.003				
CE(Long Formula) = 0.31				
T/S(PSI) 0 Y/S(PSI) 0 EL(%) 0.00 RA(%) 0.00				
Brinell Hardness 150 BHN 156 BHN				
39	400	J126	1/2 3M/6M A105 HXPL T	SA/A105
C 0.170 Cb 0.000 Co 0.000 Cr 0.080 Cu 0.080 Mn 0.640 Mo 0.020 Ni 0.050				
P 0.007 S 0.006 Si 0.230 V 0.002				
CE(Long Formula) = 0.31				
T/S(PSI) 75,667 Y/S(PSI) 0 EL(%) 0.00 RA(%) 0.00				
Brinell Hardness 150 BHN 156 BHN				
40	250	J133	3/4 3M/6M A105 HXPL T	SA/A105
C 0.140 Cb 0.000 Co 0.000 Cr 0.080 Cu 0.080 Mn 0.640 Mo 0.020 Ni 0.040				
P 0.008 S 0.005 Si 0.160 V 0.002				
CE(Long Formula) = 0.28				
T/S(PSI) 74,109 Y/S(PSI) 0 EL(%) 0.00 RA(%) 0.00				
Brinell Hardness 159 BHN 165 BHN				
41	75	5117	1 1/2 3M/6M A105 HXPL T	SA/A105
Al 0.026 C 0.210 Co 0.003 Cr 0.130 Cu 0.040 Mn 1.050 Mo 0.002 Nb 0.009				
Ni 0.020 P 0.006 S 0.022 Si 0.200 V 0.004				
CE(Long Formula) = 0.42				
T/S(PSI) 76,800 Y/S(PSI) 45,300 EL(%) 30.00 RA(%) 47.80				
Brinell Hardness 167 BHN 167 BHN				

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
42	1,050	J161	1 3M/6M A105 HXPL T	SA/A105
C 0.200 Cb 0.002 Co 0.010 Cr 0.060 Cu 0.070 Mn 0.940 Mo 0.010 Ni 0.040				
P 0.007 S 0.002 Si 0.200 V 0.002				
CE(Long Formula) = 0.38				
T/S(PSI) 82,382 Y/S(PSI) 71,500 EL(%) 22.00 RA(%) 63.00				
Brinell Hardness 180 BHN 162 BHN				
43	180	D959	2 3M/6M A105 HXPL T	SA/A105
Al 0.040 C 0.190 Co 0.006 Cr 0.210 Cu 0.260 Mn 1.020 Mo 0.020 Nb 0.003				
Ni 0.060 P 0.013 S 0.020 Si 0.224 V 0.019				
CE(Long Formula) = 0.43				
T/S(PSI) 81,500 Y/S(PSI) 50,800 EL(%) 30.00 RA(%) 64.00				
Brinell Hardness 163 BHN 164 BHN				
49	25	D930	3/4X 1/2 3000 A105 REDU T	SA/A105
Al 0.022 C 0.210 Co 0.004 Cr 0.100 Cu 0.110 Mn 1.080 Mo 0.010 Nb 0.009				
Ni 0.040 P 0.008 S 0.020 Si 0.260 V 0.004				
CE(Long Formula) = 0.42				
T/S(PSI) 79,700 Y/S(PSI) 48,000 EL(%) 35.00 RA(%) 64.00				
Brinell Hardness 170 BHN 170 BHN				
50	25	D933	1/2 3M/6M A105 RDPL T	SA/A105
Al 0.028 C 0.200 Co 0.010 Cr 0.120 Cu 0.100 Mn 0.990 Mo 0.010 Nb 0.001				
Ni 0.040 P 0.007 S 0.021 Si 0.260 V 0.004				
CE(Long Formula) = 0.40				
T/S(PSI) 77,900 Y/S(PSI) 55,100 EL(%) 30.00 RA(%) 61.00				
Brinell Hardness 148 BHN 143 BHN				

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SHIP TO: EDGEN MURRAY CORPORATION

8445 MILLER RD 2

HOUSTON TX 77049

United States

ITEM	QUANTITY	LOT NO.	GRADE OR SPECIFICATION NO.	
			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
51	25	J116	3/4 3M/6M A105 RDPL T	SA/A105
Al 0.021 C 0.190 Co 0.010 Cr 0.100 Cu 0.080 Mn 0.970 Mo 0.010 Nb 0.002				
Ni 0.040 P 0.008 S 0.023 Si 0.230 V 0.003				
CE(Long Formula) = 0.38				
T/S(PSI) 72,300 Y/S(PSI) 46,600 EL(%) 36.00 RA(%) 70.00				
Brinell Hardness 146 BHN 147 BHN				
52	100	D954	1 3M/6M A105 RDPL T	SA/A105
Al 0.029 C 0.190 Co 0.010 Cr 0.110 Cu 0.110 Mn 0.990 Mo 0.010 Nb 0.001				
Ni 0.040 P 0.008 S 0.023 Si 0.260 V 0.004				
CE(Long Formula) = 0.39				
T/S(PSI) 74,400 Y/S(PSI) 47,400 EL(%) 32.00 RA(%) 65.00				
Brinell Hardness 146 BHN 149 BHN				
53	20	51865	3/4X 3/4X 1/2 3000 A105 TEE T	SA/A105
Al 0.022 C 0.210 Co 0.004 Cr 0.100 Cu 0.110 Mn 1.080 Mo 0.010 Nb 0.009				
Ni 0.040 P 0.008 S 0.020 Si 0.260 V 0.004				
CE(Long Formula) = 0.42				
T/S(PSI) 82,588 Y/S(PSI) 52,849 EL(%) 31.70 RA(%) 66.26				
Brinell Hardness 159 BHN 159 BHN				
54	30	51796	1 X1 X 1/2 3000 A105 TEE T	SA/A105
Al 0.027 C 0.180 Co 0.000 Cr 0.130 Cu 0.140 Mn 1.010 Mo 0.030				
Ni 0.070 P 0.008 S 0.024 Si 0.220 V 0.033				
CE(Long Formula) = 0.40				
T/S(PSI) 82,412 Y/S(PSI) 56,619 EL(%) 30.45 RA(%) 64.66				
Brinell Hardness 165 BHN 162 BHN				

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by

Paige Smith
Quality Assurance Representative

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DATE 02/13/2025

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ITEM	QUANTITY	LOT NO.	GRADE OR SPECIFICATION NO.	
			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
56	20	51769	1 X1 X 3/4 3000 A105 TEE T	SA/A105
Al 0.026 C 0.170 Cb 0.001 Co 0.006 Cr 0.140 Cu 0.190 Mn 1.050 Mo 0.030				
Ni 0.080 P 0.008 S 0.022 Si 0.250 V 0.035				
CE(Long Formula) = 0.40				
T/S(PSI) 82,168 Y/S(PSI) 56,666 EL(%) 33.10 RA(%) 66.83				
Brinell Hardness 165 BHN 162 BHN				
57	15	51925	2 X 2 X 1 3000 A105 TEE T	SA/A105
Al 0.022 C 0.200 Co 0.003 Cr 0.110 Cu 0.040 Mn 1.100 Mo 0.006 Nb 0.010				
Ni 0.030 P 0.006 S 0.023 Si 0.200 V 0.004				
CE(Long Formula) = 0.41				
T/S(PSI) 80,484 Y/S(PSI) 52,124 EL(%) 33.90 RA(%) 63.62				
Brinell Hardness 147 BHN 144 BHN				
58	25	D812	1/4 3M/6M A105 SQPL T	SA/A105
C 0.170 Cb 0.000 Co 0.000 Cr 0.060 Cu 0.100 Mn 0.750 Mo 0.010 Ni 0.040				
P 0.006 S 0.006 Si 0.230 V 0.002				
CE(Long Formula) = 0.32				
T/S(PSI) 0 Y/S(PSI) 0 EL(%) 0.00 RA(%) 0.00				
Brinell Hardness 153 BHN 165 BHN				
59	125	D967	1/2 3M/6M A105 SQPL T	SA/A105
C 0.140 Cb 0.000 Co 0.000 Cr 0.080 Cu 0.080 Mn 0.640 Mo 0.020 Ni 0.040				
P 0.008 S 0.005 Si 0.160 V 0.002				
CE(Long Formula) = 0.28				
T/S(PSI) 0 Y/S(PSI) 0 EL(%) 0.00 RA(%) 0.00				
Brinell Hardness 141 BHN 156 BHN				

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by

Paige Smith

Paige Smith

Quality Assurance Representative

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ITEM	QUANTITY	LOT NO.	GRADE OR SPECIFICATION NO.	
			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
60	25	D939	3/4 3M/6M A105 SQPL T	SA/A105
C 0.170 Cb 0.000 Co 0.000 Cr 0.070 Cu 0.080 Mn 0.760 Mo 0.010 Ni 0.040				
P 0.006 S 0.002 Si 0.190 V 0.002				
CE(Long Formula) = 0.32				
T/S(Psi) 0 Y/S(Psi) 0 EL(%) 0.00 RA(%) 0.00				
Brinell Hardness 153 BHN 162 BHN				
61	25	D806	1 3M/6M A105 SQPL T	SA/A105
C 0.170 Cb 0.000 Co 0.000 Cr 0.060 Cu 0.090 Mn 0.730 Mo 0.020 Ni 0.040				
P 0.006 S 0.013 Si 0.230 V 0.002				
CE(Long Formula) = 0.32				
T/S(Psi) 0 Y/S(Psi) 0 EL(%) 0.00 RA(%) 0.00				
Brinell Hardness 150 BHN 165 BHN				
63	25	51658	1/4 3000 A105 TEE T	SA/A105
Al 0.023 C 0.190 Co 0.010 Cr 0.120 Cu 0.130 Mn 0.970 Mo 0.010 Nb 0.001				
Ni 0.040 P 0.010 S 0.020 Si 0.210 V 0.003				
CE(Long Formula) = 0.39				
T/S(Psi) 75,248 Y/S(Psi) 49,293 EL(%) 36.10 RA(%) 67.31				
Brinell Hardness 139 BHN 141 BHN				
64	25	51870	1/2 3000 A105 TEE T	SA/A105
Al 0.017 C 0.200 Co 0.004 Cr 0.080 Cu 0.140 Mn 1.050 Mo 0.009 Nb 0.009				
Ni 0.040 P 0.008 S 0.024 Si 0.210 V 0.004				
CE(Long Formula) = 0.41				
T/S(Psi) 79,971 Y/S(Psi) 54,816 EL(%) 31.55 RA(%) 59.71				
Brinell Hardness 153 BHN 156 BHN				

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ITEM	QUANTITY	LOT NO.	GRADE OR SPECIFICATION NO.	
			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
65	25	51879	1/2 6000 A105 TEE T	SA/A105
Al 0.022 C 0.170 Cb 0.000 Co 0.006 Cr 0.140 Cu 0.120 Mn 1.060 Mo 0.040				
Ni 0.080 P 0.010 S 0.023 Si 0.200 V 0.033				
CE(Long Formula) = 0.40				
T/S(PSI) 82,000 Y/S(PSI) 54,000 EL(%) 31.00 RA(%) 69.00				
Brinell Hardness 167 BHN 167 BHN				
66	40	51918	11/2 3000 A105 TEE T	SA/A105
Al 0.022 C 0.200 Co 0.003 Cr 0.110 Cu 0.040 Mn 1.080 Mo 0.013 Nb 0.010				
Ni 0.030 P 0.007 S 0.019 Si 0.240 V 0.004				
CE(Long Formula) = 0.41				
T/S(PSI) 81,186 Y/S(PSI) 56,019 EL(%) 34.45 RA(%) 61.68				
Brinell Hardness 150 BHN 156 BHN				
67	40	51984	11/2 3000 A105 TEE S	SA/A105
Al 0.026 C 0.210 Co 0.003 Cr 0.130 Cu 0.040 Mn 1.050 Mo 0.002 Nb 0.009				
Ni 0.020 P 0.006 S 0.022 Si 0.200 V 0.004				
CE(Long Formula) = 0.42				
T/S(PSI) 80,580 Y/S(PSI) 53,981 EL(%) 34.50 RA(%) 66.01				
Brinell Hardness 159 BHN 159 BHN				
68	100	51912	1 3000 A105 TEE T	SA/A105
Al 0.022 C 0.200 Co 0.003 Cr 0.110 Cu 0.040 Mn 1.100 Mo 0.006 Nb 0.010				
Ni 0.030 P 0.006 S 0.023 Si 0.200 V 0.004				
CE(Long Formula) = 0.41				
T/S(PSI) 80,484 Y/S(PSI) 52,124 EL(%) 33.90 RA(%) 63.62				
Brinell Hardness 147 BHN 144 BHN				

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
69	50	51895	1 3000 A105 TEE S	SA/A105
Al 0.025 C 0.200 Co 0.002 Cr 0.100 Cu 0.030 Mn 1.050 Mo 0.003 Nb 0.011				
Ni 0.010 P 0.004 S 0.021 Si 0.240 V 0.005				
CE(Long Formula) = 0.40				
T/S(PSI) 81,756 Y/S(PSI) 52,062 EL(%) 33.15 RA(%) 61.09				
Brinell Hardness 156 BHN 156 BHN				
70	15	51789	1 6000 A105 TEE T	SA/A105
Al 0.026 C 0.170 Cb 0.001 Co 0.006 Cr 0.150 Cu 0.210 Mn 1.020 Mo 0.030				
Ni 0.070 P 0.009 S 0.020 Si 0.240 V 0.031				
CE(Long Formula) = 0.40				
T/S(PSI) 81,484 Y/S(PSI) 56,075 EL(%) 31.55 RA(%) 61.79				
Brinell Hardness 153 BHN 153 BHN				
71	100	51890	2 3000 A105 TEE T	SA/A105
Al 0.024 C 0.220 Co 0.002 Cr 0.150 Cu 0.020 Mn 1.050 Mo 0.001 Nb 0.012				
Ni 0.010 P 0.004 S 0.018 Si 0.230 V 0.004				
CE(Long Formula) = 0.43				
T/S(PSI) 79,727 Y/S(PSI) 50,246 EL(%) 33.65 RA(%) 61.53				
Brinell Hardness 150 BHN 147 BHN				
75	18	D837	1 1/4 3000 A105 UNAR T	SA/A105
		MALE		
Al 0.025 C 0.200 Co 0.003 Cr 0.130 Cu 0.050 Mn 1.050 Mo 0.007 Nb 0.009				
Ni 0.020 P 0.007 S 0.021 Si 0.230 V 0.004				
CE(Long Formula) = 0.41				
T/S(PSI) 81,800 Y/S(PSI) 50,900 EL(%) 32.00 RA(%) 52.60				
Brinell Hardness 166 BHN 166 BHN				

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ITEM	QUANTITY	LOT NO.	GRADE OR SPECIFICATION NO.	
			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
		D947	11/4 3000 A105 UNAR T	SA/A105
		FEMALE	Al 0.023 C 0.190 Co 0.002 Cr 0.120 Cu 0.020 Mn 1.060 Mo 0.001 Nb 0.012 Ni 0.010 P 0.004 S 0.021 Si 0.200 V 0.004 CE(Long Formula) = 0.39 T/S(Psi) 76,100 Y/S(Psi) 44,600 EL(%) 32.00 RA(%) 60.40 Brinell Hardness 154 BHN 155 BHN	
		H4HI	11/4 3000 A105 UNAR T	SA/A105
		NUT	Al 0.024 C 0.170 Cb 0.001 Co 0.005 Cr 0.120 Cu 0.200 Mn 1.060 Mo 0.030 Ni 0.090 P 0.006 S 0.018 Si 0.250 V 0.031 CE(Long Formula) = 0.40 T/S(Psi) 81,645 Y/S(Psi) 56,921 EL(%) 31.05 RA(%) 61.30 Brinell Hardness 162 BHN 162 BHN	
77	250	D900	1 3000 A105 UNAR T	SA/A105
		MALE	Al 0.020 C 0.200 Co 0.002 Cr 0.120 Cu 0.030 Mn 1.060 Mo 0.003 Nb 0.010 Ni 0.010 P 0.006 S 0.019 Si 0.230 V 0.004 CE(Long Formula) = 0.40 T/S(Psi) 80,700 Y/S(Psi) 50,900 EL(%) 34.00 RA(%) 60.80 Brinell Hardness 169 BHN 169 BHN	
		J102	1 3000 A105 UNAR T	SA/A105
		FEMALE	Al 0.021 C 0.190 Co 0.006 Cr 0.130 Cu 0.230 Mn 0.980 Mo 0.020 Nb 0.002 Ni 0.090 P 0.009 S 0.017 Si 0.158 V 0.027 CE(Long Formula) = 0.41 T/S(Psi) 78,300 Y/S(Psi) 48,800 EL(%) 26.00 RA(%) 48.00 Brinell Hardness 159 BHN 155 BHN	

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
		H3DC	1 3000 A105 UNAR T	SA/A105
		NUT	Al 0.023 C 0.170 Cb 0.000 Co 0.006 Cr 0.100 Cu 0.120 Mn 1.130 Mo 0.020 Ni 0.050 P 0.007 S 0.021 Si 0.240 V 0.035 CE(Long Formula) = 0.40 T/S(PSI) 80,045 Y/S(PSI) 58,777 EL(%) 32.55 RA(%) 60.75 Brinell Hardness 153 BHN 153 BHN	
79	36	J142	1 6000 A105 UNAR T	SA/A105
		MALE	Al 0.026 C 0.190 Co 0.002 Cr 0.100 Cu 0.030 Mn 1.080 Mo 0.004 Nb 0.009 Ni 0.020 P 0.006 S 0.024 Si 0.220 V 0.004 CE(Long Formula) = 0.39 T/S(PSI) 75,200 Y/S(PSI) 43,600 EL(%) 30.00 RA(%) 57.00 Brinell Hardness 156 BHN 156 BHN	
		D908	1 6000 A105 UNAR T	SA/A105
		FEMALE	Al 0.024 C 0.200 Co 0.002 Cr 0.120 Cu 0.030 Mn 1.090 Mo 0.001 Nb 0.011 Ni 0.020 P 0.008 S 0.022 Si 0.200 V 0.004 CE(Long Formula) = 0.41 T/S(PSI) 77,200 Y/S(PSI) 44,400 EL(%) 34.00 RA(%) 64.20 Brinell Hardness 163 BHN 156 BHN	
		D885	1 6000 A105 UNAR T	SA/A105
		NUT	Al 0.028 C 0.170 Cb 0.003 Co 0.006 Cr 0.100 Cu 0.160 Mn 1.170 Mo 0.030 Ni 0.070 P 0.005 S 0.017 Si 0.230 V 0.032 CE(Long Formula) = 0.41 T/S(PSI) 78,800 Y/S(PSI) 52,963 EL(%) 29.00 RA(%) 59.00 Brinell Hardness 159 BHN 158 BHN	

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
44	40	D914	3/4 3M/6M A105N HXPL T	SA/A105N
Al 0.026 C 0.200 Co 0.012 Cr 0.070 Cu 0.300 Mn 1.050 Mo 0.020 Nb 0.005				
Ni 0.090 P 0.009 S 0.023 Si 0.233 V 0.022				
CE(Long Formula) = 0.42				
T/S(PSI) 76,474 Y/S(PSI) 53,609 EL(%) 34.20 RA(%) 60.48				
Brinell Hardness 137 BHN 137 BHN				
45	10	D817	1 3M/6M A105N HXPL T	SA/A105N
C 0.210 Co 0.010 Cr 0.060 Cu 0.070 Mn 0.940 Mo 0.020 Nb 0.001 Ni 0.030				
P 0.007 S 0.001 Si 0.190 V 0.003				
CE(Long Formula) = 0.39				
T/S(PSI) 73,000 Y/S(PSI) 59,500 EL(%) 29.00 RA(%) 64.00				
Brinell Hardness 153 BHN 153 BHN				
13	15	51703	11/4 3000 LF2 90EL S	SA/A350 LF2 CL1
Al 0.025 C 0.210 Co 0.002 Cr 0.130 Cu 0.030 Mn 1.070 Mo 0.004 Nb 0.011				
Ni 0.020 P 0.007 S 0.018 Si 0.280 V 0.004				
CE(Long Formula) = 0.42				
T/S(PSI) 77,388 Y/S(PSI) 50,772 EL(%) 33.95 RA(%) 62.85				
Brinell Hardness 139 BHN 137 BHN				
Charpy -50 F 61/42/33 (Ft-Lbs)				
Average 45.30				
		51591	11/4 3000 LF2 90EL S	SA/A350 LF2 CL1
Al 0.021 C 0.180 Co 0.001 Cr 0.100 Cu 0.170 Mn 1.150 Mo 0.020				
Ni 0.060 P 0.008 S 0.017 Si 0.250 V 0.032				
CE(Long Formula) = 0.42				
T/S(PSI) 79,326 Y/S(PSI) 55,812 EL(%) 34.50 RA(%) 63.29				
Brinell Hardness 141 BHN 141 BHN				
Charpy -50 F 74/83/78 (Ft-Lbs)				
Average 78.30				

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
15	20	51917	1 1/2 3000 LF2 90EL T	SA/A350 LF2 CL1
			Al 0.028 C 0.200 Co 0.003 Cr 0.090 Cu 0.050 Mn 1.040 Mo 0.024 Nb 0.010	
			Ni 0.030 P 0.005 S 0.021 Si 0.230 V 0.004	
			CE(Long Formula) = 0.40	
			T/S(PSI) 73,800 Y/S(PSI) 48,317 EL(%) 36.30 RA(%) 62.32	
			Brinell Hardness 137 BHN 137 BHN	
			Charpy -50 F 54/27/43 (Ft-Lbs)	
			Average 41.30	
27	70	D993	1 3M LF2 COUP S	SA/A350 LF2 CL1
			Al 0.029 C 0.180 Co 0.006 Cr 0.070 Cu 0.260 Mn 1.040 Mo 0.020 Nb 0.001	
			Ni 0.080 P 0.004 S 0.025 Si 0.215 V 0.025	
			CE(Long Formula) = 0.40	
			T/S(PSI) 74,639 Y/S(PSI) 52,386 EL(%) 34.80 RA(%) 63.57	
			Brinell Hardness 141 BHN 139 BHN	
			Charpy -50 F 55/105/77 (Ft-Lbs)	
			Average 79.00	
46	70	D592	1/2 3M/6M LF2 HXPL T	SA/A350 LF2 CL1
			Al 0.023 C 0.190 Co 0.010 Cr 0.120 Cu 0.110 Mn 0.990 Mo 0.020 Nb 0.002	
			Ni 0.050 P 0.008 S 0.022 Si 0.240 V 0.002	
			CE(Long Formula) = 0.39	
			T/S(PSI) 73,017 Y/S(PSI) 47,962 EL(%) 35.70 RA(%) 66.86	
			Brinell Hardness 137 BHN 137 BHN	
			Charpy -50 F 82/35/38 (Ft-Lbs)	
			Average 51.70	

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
47	65	D894	1 3M/6M LF2 HXPL T	SA/A350 LF2 CL1
Al 0.023 C 0.210 Co 0.002 Cr 0.120 Cu 0.040 Mn 1.050 Mo 0.003 Nb 0.009				
Ni 0.020 P 0.007 S 0.024 Si 0.220 V 0.004				
CE(Long Formula) = 0.41				
T/S(PSI) 76,121 Y/S(PSI) 49,885 EL(%) 35.05 RA(%) 60.88				
Brinell Hardness 137 BHN 137 BHN				
Charpy -50 F 55/65/34 (Ft-Lbs)				
Average 51.30				
48	40	D463	2 3M/6M LF2 HXPL T	SA/A350 LF2 CL1
C 0.190 Cb 0.012 Co 0.004 Cr 0.120 Cu 0.050 Mn 1.070 Mo 0.003 Ni 0.020				
P 0.006 S 0.020 Si 0.230 V 0.004				
CE(Long Formula) = 0.40				
T/S(PSI) 75,940 Y/S(PSI) 50,351 EL(%) 36.30 RA(%) 61.61				
Brinell Hardness 137 BHN 137 BHN				
Charpy -50 F 44/29/48 (Ft-Lbs)				
Average 40.30				
72	17	51918	11/2 3000 LF2 TEE T	SA/A350 LF2 CL1
Al 0.022 C 0.200 Co 0.003 Cr 0.110 Cu 0.040 Mn 1.080 Mo 0.013 Nb 0.010				
Ni 0.030 P 0.007 S 0.019 Si 0.240 V 0.004				
CE(Long Formula) = 0.41				
T/S(PSI) 74,494 Y/S(PSI) 49,931 EL(%) 36.00 RA(%) 66.11				
Brinell Hardness 139 BHN 139 BHN				
Charpy -50 F 112/109/93 (Ft-Lbs)				
Average 104.70				

We certify that the data on this sheet is a true copy taken from our records of material furnished us by the production mill, or as obtained by additional laboratory checks.

by

Paige Smith

Paige Smith

Quality Assurance Representative

CMTR: Rev 3

CERTIFIED MILL TEST REPORT



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(814) 542-2545 • (800) 345-7546 • FAX (814) 542-4906

www.bonneyforge.com

CUSTOMER: EDGEN MURRAY CORPORATION

DATE 02/13/2025

CUSTOMER

BONNEY ORDER NO. B000358755

ORDER NO.: DIS-313586

SHIP TO: EDGEN MURRAY CORPORATION

8445 MILLER RD 2

HOUSTON TX 77049

United States

ITEM	QUANTITY	LOT NO.	GRADE OR SPECIFICATION NO. CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:
72	3	51907	11/2 3000 LF2 TEE T SA/A350 LF2 CL1 Al 0.038 C 0.200 Co 0.006 Cr 0.200 Cu 0.250 Mn 0.950 Mo 0.020 Nb 0.004 Ni 0.070 P 0.008 S 0.025 Si 0.263 V 0.021 CE(Long Formula) = 0.43 T/S(PSI) 75,866 Y/S(PSI) 51,020 EL(%) 35.60 RA(%) 63.90 Brinell Hardness 137 BHN 137 BHN Charpy -50 F 125/101/137 (Ft-Lbs) Average 121.00
73	10	51602	2 3000 LF2 TEE T SA/A350 LF2 CL1 Al 0.027 C 0.180 Co 0.006 Cr 0.170 Cu 0.260 Mn 1.010 Mo 0.020 Nb 0.001 Ni 0.070 P 0.010 S 0.025 Si 0.172 V 0.022 CE(Long Formula) = 0.41 T/S(PSI) 72,930 Y/S(PSI) 48,989 EL(%) 37.50 RA(%) 64.47 Brinell Hardness 137 BHN 137 BHN Charpy -50 F 34/34/23 (Ft-Lbs) Average 30.30
81	25	D947 MALE	11/2 3M LF2 UNAR T SA/A350 LF2 CL1 Al 0.023 C 0.190 Co 0.002 Cr 0.120 Cu 0.020 Mn 1.060 Mo 0.001 Nb 0.012 Ni 0.010 P 0.004 S 0.021 Si 0.200 V 0.004 CE(Long Formula) = 0.39 T/S(PSI) 73,445 Y/S(PSI) 49,696 EL(%) 36.25 RA(%) 61.68 Brinell Hardness 137 BHN 137 BHN Charpy -50 F 37/48/60 (Ft-Lbs) Average 48.30

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			CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:	
D525		11/2 3M LF2 UNAR T	SA/A350 LF2 CL1	
FEMALE		C 0.180 Cb 0.000 Co 0.006 Cr 0.110 Cu 0.190 Mn 1.040 Mo 0.020 Ni 0.070		
		P 0.008 S 0.017 Si 0.260 V 0.032		
		CE(Long Formula) = 0.40		
		T/S(PSI) 76,517 Y/S(PSI) 54,895 EL(%) 34.20 RA(%) 58.13		
		Brinell Hardness 137 BHN 139 BHN		
		Charpy -50 F 57/73/74 (Ft-Lbs)		
		Average 68.00		
H5CC		11/2 3M LF2 UNAR T	SA/A350 LF2 CL1	
NUT		Al 0.033 C 0.170 Cb 0.000 Co 0.006 Cr 0.100 Cu 0.190 Mn 1.130 Mo 0.020		
		Ni 0.070 P 0.009 S 0.023 Si 0.240 Sn 0.007 V 0.035		
		CE(Long Formula) = 0.41		
		T/S(PSI) 75,850 Y/S(PSI) 53,778 EL(%) 37.50 RA(%) 62.37		
		Brinell Hardness 137 BHN 137 BHN		
		Charpy -50 F 92/104/114 (Ft-Lbs)		
		Average 103.30		

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ITEM QUANTITY LOT NO.

GRADE OR SPECIFICATION NO.
CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:

1. THE FITTINGS SUPPLIED ARE IN ACCORDANCE WITH PURCHASE ORDER SPECIFICATIONS.
2. THE PRODUCT SUPPLIED WAS INSPECTED IN ACCORDANCE WITH EN 10204:2004 EDITION TYPE 3.1 INSPECTION DOCUMENT. (EUROPEAN STANDARD)
3. ELONGATION TEST RESULTS ARE OBTAINED USING STANDARD ROUND SPECIMEN, 2 INCH OR 50 MM GAGE LENGTH.
4. ITEM - 1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 16, 17, 19, 20, 22, 23, 24, 25, 28, 30, 31, 32, 33, 34, 35, 37, 38, 39, 40, 41, 42, 43, 44, 45, 49, 50, 51, 52, 53, 54, 56, 57, 58, 59, 60, 61, 63, 64, 65, 66, 67, 68, 69, 70, 71, 75, 77, 79 - CERTIFYING ASTM A105-24 / ASME SA105-23 EDITION.
5. ITEM - 1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 16, 17, 19, 20, 22, 23, 24, 25, 28, 30, 31, 32, 33, 34, 35, 37, 38, 39, 40, 41, 42, 43, 49, 50, 51, 52, 53, 54, 56, 57, 58, 59, 60, 61, 63, 64, 65, 66, 67, 68, 69, 70, 71, 75, 77, 79 - THE MATERIAL SUPPLIED MEETS THE REQUIREMENTS OF NACE MR0175/ISO 15156-2 - 2015 EDITION.
6. ITEM - 13, 15, 27, 46, 47, 48, 72, 73, 81 - CERTIFYING TO ASTM A350 LF2 24 REVISION / ASME SA350 LF2 23 EDITION.
7. ITEM - 13, 15, 27, 46, 47, 48, 72, 73, 81 - THE MATERIAL OF THE FITTINGS SUPPLIED IS ASTM A350-LF2 BUT MEETS THE REQUIREMENTS OF ASME SA350-LF2.
8. ITEM - 13, 15, 27, 46, 47, 48, 72, 73, 81 - THE CHARPY V-NOTCH IMPACT TEST WAS PERFORMED AT -50 DEGREES FAHRENHEIT IN ACCORDANCE WITH A350-LF2.
9. ITEM - 13, 15, 27, 46, 47, 48, 72, 73, 81 - THE MATERIAL SUPPLIED AS A350 LF2 CL1 MEETS THE REQUIREMENTS OF BOTH NACE MRO103/ISO 17945 - 2015 & NACE MRO175/ISO 15156-2 - 2015 EDITION.
10. ITEM - 13, 15, 27, 46, 47, 48, 72, 73, 81 - THE MATERIAL SUPPLIED WAS NORMALIZED IN ACCORDANCE WITH ASTM A350 HEAT TREATING REQUIREMENTS.
11. ITEM - 37, 38, 39, 40, 58, 59, 60, 61 - THE MATERIAL SUPPLIED IDENTIFIED AS SA/A105-QT WAS QUENCHED AND TEMPERED IN ACCORDANCE WITH ASTM A105 HEAT TREATING REQUIREMENTS.
12. ITEM - 44, 45 - THE MATERIAL SUPPLIED IDENTIFIED AS A105N WAS NORMALIZED IN ACCORDANCE WITH ASTM A105 HEAT TREATING REQUIREMENTS.

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by

Paige Smith
Quality Assurance Representative

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Desc Forged Equal Tee ASTM A/SA105 NPT Threaded,2" 3000#

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8445 MILLER RD 2
HOUSTON TX 77049
United States

ITEM	QUANTITY	LOT NO.
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GRADE OR SPECIFICATION NO. CHEMICAL ANALYSIS, PHYSICAL PROPERTIES, REMARKS:
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13. ITEM - 44, 45 -THE MATERIAL SUPPLIED AS A105N MEETS THE REQUIREMENTS OF BOTH NACE MR0103/ISO 17945 - 2015 & NACE MR0175/ISO 15156-2 - 2015 EDITION.

14. ITEM - 42 -THE MATERIAL SUPPLIED WAS NORMALIZED IN ACCORDANCE WITH ASTM A105 HEAT TREATING REQUIREMENTS.
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